

6-Test scenarios

Scenario A:

PID

Arrival Time

Burst Time

Quantum(RR)

4

Add Process

Delete

Reset

PID	Arrival	Burst
1	0	8
2	1	4
3	2	9
4	3	5

Excute

SJF Output

GantChart: 0-P1-1-P2-5-P4-10-P1-17-P3-26
Average Waiting Time = 6.50
Average TAT = 13.00
Average RT = 4.25

Round Robin Output

GantChart: 0-P1-4-P2-8-P3-12-P4-16-P1-20-P3-24-P4-25-P3-26
Average Waiting Time = 11.75
Average TAT = 18.25
Average RT = 4.50

Goal:

The goal is to calculate and compare the efficiency of each algorithm using a specific set of processes based on their arrival time and burst time.

Inputs

<i>PID</i>	<i>Arrival Time</i>	<i>Burst Time</i>
1	0	8
2	1	4
3	2	9
4	3	5

Shortest Job First (SJF) Results:

- Gantt Chart: 0-p1-1-p2-5-p4-10-p1-17-p3-26
- Average Waiting Time: 6.50
- Average Turnaround Time (TAT): 13.00
- Average Response Time (RT): 4.25

Round Robin(RR) Results:

Quantum(RR):4

- Gantt Chart: 0-p1-4-p2-8-p3-12-p4-16-p1-20-p3-24-p4-25-p3-26
- Average Waiting Time: 11.75
- Average Turnaround Time (TAT): 18.25
- Average Response Time (RT): 4.50

Conclusion

For this specific set of processes, SJF is the more effective algorithm as it minimises the overall wait time and speeds up process completion.

Scenario B:

PID	Arrival Time	Burst Time	Quantum(RR)
			2
Add ProcessDeleteResetExcute			

PID	Arrival	Burst
1	0	2
2	1	1
3	2	2
4	3	1

SJF Output

GantChart: 0-P1-2-P2-3-P4-4-P3-6
Average Waiting Time = 0.75
Average TAT = 2.25
Average RT = 0.75

Round Robin Output

GantChart: 0-P1-2-P2-3-P3-5-P4-6
Average Waiting Time = 1.00
Average TAT = 2.50
Average RT = 1.00

Goal:

The goal of this scenario is to evaluate how each algorithm handles a short-job-heavy workload.

Input:

<i>PID</i>	<i>Arrival Time</i>	<i>Burst Time</i>
1	0	2
2	1	1
3	2	2
4	3	1

SJF Results:

GantChart:0-p1-2-p2-3-p4-4-p3-6

Average Waiting Time=0.75

Average TAT=2.25

Average RT=0.75

Round Robin Results:

Quantum(RR):2

GantChart:0-p1-2-p2-3-p3-5-p4-6

Average Waiting Time=1.0

Average TAT=2.50

Average RT=1.0

Conclusion

In this short-job-heavy workload, both SJF and Round Robin produced similar performance results because all processes had relatively short burst times. The waiting time, turnaround time, and response time were all low in both algorithms.

However, SJF still demonstrated its efficiency by always selecting the shortest available process first, while Round Robin ensured fairness by distributing CPU time equally among processes

Scenario C:

PID	Arrival Time	Burst Time	Quantum(RR)
			4
Add Process Delete Reset Execute			

PID	Arrival	Burst
1	0	6
2	0	7
3	0	5
4	0	10

SJF Output

```
GantChart: 0-P3-5-P1-11-P2-18-P4-28
Average Waiting Time = 8.50
Average TAT = 15.50
Average RT = 8.50
```

Round Robin Output

```
GantChart: 0-P1-4-P2-8-P3-12-P4-16-P1-18-P2-21-P3-22-P4-26-P4-28
Average Waiting Time = 15.25
Average TAT = 22.25
Average RT = 6.00
```

Goal:

This scenario analyses the performance of Shortest Job First (SJF) and Round Robin (RR) algorithms when multiple processes arrive at the same time. The objective is to evaluate how each algorithm handles equal priority tasks and its impact on the average waiting and turnaround times.

Input Data:

<i>PID</i>	<i>Arrival Time</i>	<i>Burst Time</i>
1	0	6
2	0	7
3	0	5
4	0	10

Shortest Job First (SJF) Results:

- GantChart: 0-P3-5-P1-11-P2-18-P4-28
- Average Waiting Time = 8.50
- Average TAT = 15.50
- Average RT = 8.50

Round Robin (RR) Results:

Quantum(RR):4

- GantChart: 0-P1-4-P2-8-P3-12-P4-16-P1-18-P2-21-P3-22-P4-26-P4-28
- Average Waiting Time = 15.25
- Average TAT = 22.25
- Average RT = 6.00

Conclusion

In this scenario, all processes arrived at the same time, but they had different burst times. SJF achieved better performance in terms of Average Waiting Time and Average Turnaround Time because it prioritised the shorter processes first.

On the other hand, Round Robin distributed CPU time fairly among all processes using a fixed quantum, which improved responsiveness and reduced the Average Response Time compared to SJF. Each process was able to start execution earlier instead of waiting for shorter processes to fully complete.

Therefore, SJF was more efficient for minimising waiting and turnaround times, while Round Robin provided better fairness and responsiveness among competing processes.

Scenario D:

PID

Arrival Time

Burst Time

Quantum(RR)

3

Add Process

Delete

Reset

PID	Arrival	Burst
1	0	15
2	2	2
3	4	2
4	6	2

Excute

SJF Output

GantChart: 0-P1-2-P2-4-P3-6-P4-8-P1-21
Average Waiting Time = 1.50
Average TAT = 6.75
Average RT = 0.00

Round Robin Output

GantChart: 0-P1-3-P2-5-P1-8-P3-10-P4-12-P1-15-P1-18-P1-21
Average Waiting Time = 3.75
Average TAT = 9.00
Average RT = 2.25

PID

Arrival Time

Burst Time

Quantum(RR)

6

Add Process

Delete

Reset

PID	Arrival	Burst
1	0	15
2	2	2
3	4	2
4	6	2

Excute

SJF Output

GantChart: 0-P1-2-P2-4-P3-6-P4-8-P1-21
Average Waiting Time = 1.50
Average TAT = 6.75
Average RT = 0.00

Round Robin Output

GantChart: 0-P1-6-P2-8-P3-10-P4-12-P1-18-P1-21
Average Waiting Time = 4.50
Average TAT = 9.75
Average RT = 3.00

Goal:

The goal is to analyse the performance and stability of each algorithm when a single, high-demand process competes for resources against several smaller, incoming tasks.

Input:

<i>PID</i>	<i>Arrival Time</i>	<i>Burst Time</i>
1	0	15
2	2	2
3	4	2
4	6	2

SJF Result:

GantChart: 0-p1-2-p2-4-p3-6-p4-8-p1-21

Average Waiting Time = 1.50

Average TAT = 6.75

Average RT = 0.00

Round Robin Results:**Quantum(RR): 3**

GantChart: 0-p1-3-p2-5-p1-8-p3-10-p4-12-p1-15-p1-18-p1-21

Average Waiting Time = 3.75

Average TAT = 9.00

Average RT = 2.25

Quantum(RR): 6

GantChart: 0-P1-6-P2-8-P3-10-P4-12-P1-18-P1-21

Average Waiting Time = 4.50

Average TAT = 9.75

Average RT = 3.00

Conclusion:

SJF is the most efficient choice for this scenario.

It achieved a much lower Average Waiting Time (1.50) compared to 3.75 in Round Robin.

SJF also showed superior performance in Turnaround Time (6.75) and Response Time (0.00).

Scenario E:

PID: Arrival Time: Burst Time: Quantum(RR):

Add Process Delete Reset

PID	Arrival	Burst
1	0	8
2	1	4
3	2	9

Message



Quantum must be a positive integer greater than 0!

OK

SJF Output

Round Robin Output

<i>PID</i>	<i>Arrival Time</i>	<i>Burst Time</i>
1	0	8
2	1	4
3	2	9

Quantum Time=-5

Note: Quantum Time must be a positive integer.

PID:
 Arrival Time:
 Burst Time:
 Quantum(RR):

PID	Arrival	Burst
1	0	8
2	1	4

Message ×

Arrival Time cannot be negative!

SJF Output

Round Robin Output

<i>PID</i>	<i>Arrival Time</i>	<i>Burst Time</i>
1	0	8
2	1	4
3	-1	9

Note: Arrival Time must be a positive integer.

PID

Arrival Time

Burst Time

Quantum(RR)

Add Process

Delete

Reset

PID	Arrival	Burst
1	0	8
2	1	4
3	2	9

Message

Please enter a quantum value!

OK

SJF Output

Round Robin Output

PID

Arrival Time

Burst Time

Quantum(RR)

3

9

Add Process

Delete

Reset

PID	Arrival	Burst
1	0	8
2	1	4

Message

Error: Missing required fields!

OK

SJF Output

Round Robin Output

PID:
 Arrival Time:
 Burst Time:
 Quantum(RR):

Add Process Delete Reset

PID	Arrival	Burst
1	0	8
2	1	4

Message

 Duplicate Process ID detected!

OK

SJF Output

Round Robin Output

<i>PID</i>	<i>Arrival Time</i>	<i>Burst Time</i>
1	0	8
2	1	4
1	2	9

Note: process ID must be unique.